Martin Raič 5.02

računski: pisni	2 kolokvija
teoretični: ustni	
projektna naloga	

1. Centralni limitni izrek

$$S_n \sim Bin(n, \tfrac{1}{2}) \qquad P(S_n = k) = \binom{n}{k} \cdot 2^{-n}$$

$$\frac{P(S_n = k+1)}{P(S_n = k)} = \frac{\binom{n}{k+1} 2^{-n}}{\binom{n}{k} 2^{-n}} = \frac{\cancel{n!}\,(n-k)!\,k!}{(k+1)!\,(n-k-1)!\,\cancel{n!}} = \frac{n-k}{k+1}$$

Naj bo n sod $n = 2m$

$P(S_n = k)$ je maksimalna pri $k = m$ DN

(gledamo razmerje $\frac{P(S_n = k+1)}{P(S_n = k)}$)

$d = 0, 1, \ldots$

$$\frac{P(S_{2m} = m+d)}{P(S_{2m} = m)} = \frac{P(S_{2m} = m+1)\, P(S_{2m} = m+2) \cdots P(S_{2m} = m+d)}{P(S_{2m} = m)\, P(S_{2m} = m+1) \cdots} =$$

$$= \frac{2m-m}{m+1} \cdot \frac{2m-m-1}{m+2} \cdots \frac{2m-m-d+1}{m+d} =$$

$$= \frac{m \cdot (m-1) \cdot (m-2) \cdots (m-d+1)}{(m+1) \cdot \cdots \cdots (m+d)} =$$

$$= \left(\frac{1}{1+\frac{1}{m}}\right)\left(\frac{1-\frac{1}{m}}{1+\frac{2}{m}}\right) \cdots \left(\frac{1-\frac{d-1}{m}}{1+\frac{d}{m}}\right) \approx$$

$$\approx e^{-\frac{1}{m} - \frac{2}{m} - \cdots - \frac{d-1}{m} - \frac{1}{m} - \frac{2}{m} \cdots - \frac{d}{m}} = e^{-\frac{d(d-1)}{2} - \frac{d(d+1)}{2}}$$

$$\ln(1+x) \approx x \qquad 1 + x \approx e^x \qquad = e^{-\frac{d^2}{m}}$$

To velja tudi za $d = -1, -2, \ldots$ zaradi sodosti

$|d| \ll m$: $P(S_{2m} = m+d) \approx P(S_{2m} = m) \cdot e^{-\frac{d^2}{m}}$

Izkaže se: $$\lim_{m \to \infty} \sum_{d=-\infty}^{\infty} \left| \frac{P(S_{2m} = m+d)}{P(S_{2m} = m)} - e^{-\frac{d^2}{m}} \right| = 0$$

Izkaže se: $\lim_{m\to\infty} \sum_{d=-\infty}^{\infty} \left| \frac{P(S_{2m}=m+d)}{P(S_{2m}=m)} - e^{-\frac{d^2}{m}} \right| = 0$

seštevek verjetnosti

$$\sum_{d=-\infty}^{\infty} \frac{P(S_{2m}=m+d)}{P(S_{2m}=m)} = \frac{1}{P(S_{2m}=m)} \approx \sum_{d=-\infty}^{\infty} e^{-\frac{d^2}{m}}$$

$$\approx \int_{-\infty}^{\infty} e^{-\frac{x^2}{m}}\,dx = \sqrt{m}\int_{-\infty}^{\infty} e^{-t^2}\,dt = \sqrt{m\pi}$$

$x = t\sqrt{m}$

$dx = \sqrt{m}\,dt$

$$\Rightarrow P(S_{2m}=m) \approx \frac{1}{\sqrt{m\pi}}$$

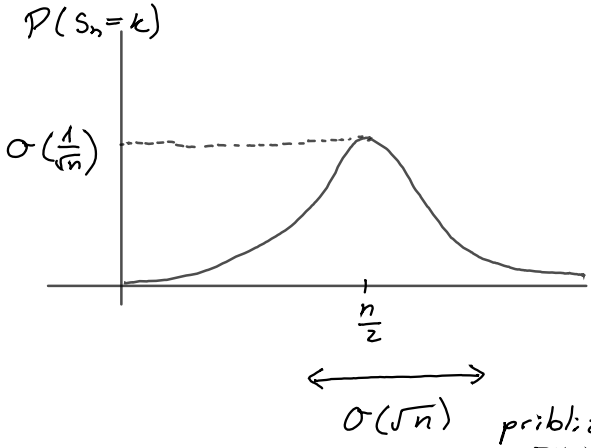
$$P(S_{2m}=m+d) \approx \frac{1}{\sqrt{m\pi}} e^{-\frac{d^2}{m}}$$

$d = k - m = k - \frac{n}{2}$:

$$\boxed{P(S_n=k) \approx \frac{1}{\sqrt{2\pi n}} e^{-\frac{2(k-\frac{n}{2})^2}{n}}}$$

De Moivreova lokalna formula

Izkaže se, da to velja tudi za lihe n



približna širina zvona

$$P(a \le S_n \le b) = \sum_{k=a}^{b} P(S_n=k) \approx \sqrt{\frac{2}{n\pi}} \sum_{k=a}^{b} e^{-\frac{2(k-\frac{n}{2})^2}{n}} \approx$$

$$\approx \sqrt{\frac{2}{n\pi}} \sum_{k=a}^{b} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} e^{-\frac{2(x-\frac{n}{2})^2}{n}} dx = \sqrt{\frac{2}{n\pi}} \int_{a-\frac{1}{2}}^{b+\frac{1}{2}} e^{-\frac{2(x-\frac{n}{2})^2}{n}} dx$$

$a, b \in \mathbb{Z} + \frac{1}{2} \quad a \le b$

$$P(a \le S_n \le b) = P(a < S_n < b) = P(a+\tfrac{1}{2} \le S_n \le b-\tfrac{1}{2}) =$$

$$= \sqrt{\frac{2}{n\pi}} \int_a^b e^{-\frac{2(x-\frac{n}{2})^2}{n}}$$

To velja tudi za splošna $a, b \in \mathbb{R}$ $a \le b$

Dobljeni približek za $P(S_n=k)$ je gostota porazdelitve $N(\frac{n}{2}, \frac{n}{4})$ v točki k.

pričakovana vrednost $\quad \sigma^2$ varianca

Podobno velja tudi za intervalske verjetnosti

Opazimo: $S \sim Bin(n,p)$ $E(S)=np$ $var(S)=np(1-p)$

Torej za $S \sim Bin(n, \frac{1}{2})$ velja $E(S)=\frac{n}{p}$ $var(S)=\frac{n}{4}$, torej enako kot za normalno porazdelitev

Binomska porazdelitev ko gre n proti neskončno je taka normalna porazdelitev, da ima isto pričakovano vrednost in varianco

Binomska porazdelitev $Bin(n,p)$
nastane kot porazdelitev vsote indikatorjev
$I_1 + \dots + I_n$ kjer so $I_1, \dots, I_n$ neodvisne
med sabo in $I_i \sim Be(p) = \begin{pmatrix} 0 & p \\ 1-p & 0 \end{pmatrix}$

Vsota veliko neodvisnih slučajnih spremenljivk z
Bernullijevo porazdelitvijo $Be(\frac{1}{2})$ je torej porazdeljena
približno normalno z ustrezno pričakovano
vrednostjo in varianco.

Primer na powerpointu
$X_1, \dots, X_n$ neodvisne. $\sim \begin{pmatrix} -1 & 0 & 9 \\ 0,5 & 0,45 & 0,05 \end{pmatrix}$

z vrjetnostjo
5% zadenemo
10 € (9 € profita)

$X_1, X_2, \dots, X_n$ neodvisne enakoporazdeljene
$E(X_1) = \mu_1 \quad E(X_1^2) < \infty \quad var(X_1) = \sigma_1^2 \quad \sigma_1 > 0$
Ce definiramo $S = X_1 + \dots + X_n$ se izkaže
$$P(a \leq S_n \leq b) \approx \frac{1}{\sigma_1 \sqrt{2\pi n}} \int_a^b e^{-\frac{(x - n\mu_1)^2}{2}} dx$$